

Source code:

```
void servo_neg90() ;

int i;

for(i = 0; i < 50; i++) {

    PORTB.F0 = 1;

    delay_us(800);
    PORTB.F0 = 0;
    delay_us(19200);
}
}

void servo0() {
    int i;
    for(i = 0; i < 50; i++) {
        PORTB.F0 = 1;
        delay_us(1500);
        PORTB.F0 = 0;
        delay_us(18500);
    }
}

void servo_pos90() {
    int i;
    for(i = 0; i < 50; i++) {
        PORTB.F0 = 1;
        delay_us(2200);
        PORTB.F0 = 0;
        delay_us(17800);
    }
}

void main() {

    ADCON1 = 0x06;

    TRISB.F0 = 0;
    TRISD.F0 = 1;

    PORTB.F0 = 0;
    servo0();

    while(1) {
```

```

if (RD0_bit == 0) {

    delay_ms(50);

    if (RD0_bit == 0) {

        while(RD0_bit == 0);

        position = position + dir;

        if(position >= 2) {
            position = 2;
            dir = -1;
        }

        if(position <= 0) {
            position = 0;
            dir = 1;
        }

    }

}

```

Output:

